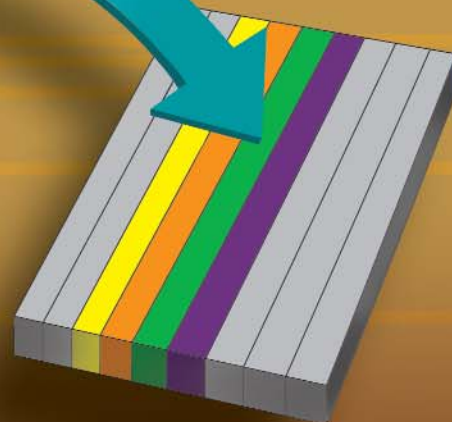
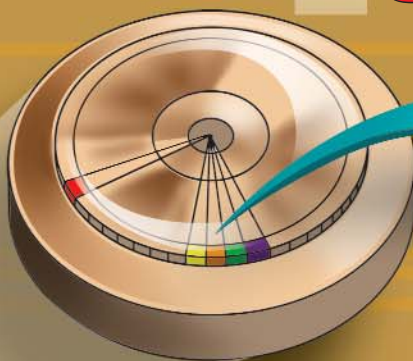


How Disk Defragmentation Works

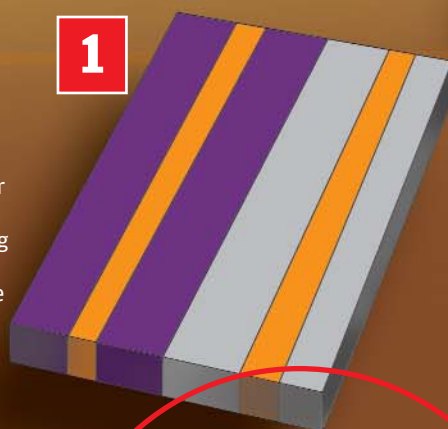
One platter of a hard disk expanded here to show a sector, the smallest unit of storage (in red). Several sectors taken together (in yellow, orange, green and blue) form a cluster, the smallest unit addressed by the filesystem.



A 2D view of the cluster

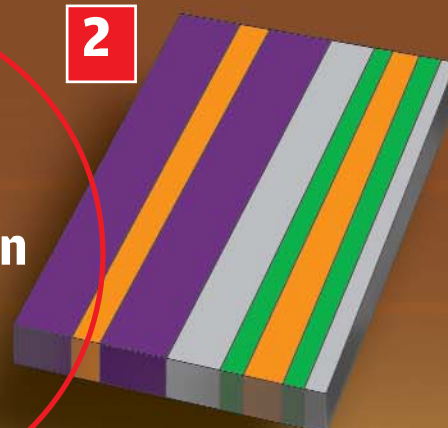
1

1. When you freshly install your OS, your files are arranged nicely, starting at the beginning of the drive. The blue and orange areas are occupied. The grey area is free space.



2

2. When you add a file (represented by the green space above) the OS places it wherever it finds space. The OS could have placed it contiguously, but it didn't.



Fragmentation

3

3. Over time, this is what your hard disk begins to look like, with files scattered all over the place, placed non-contiguously.

